Parameters We selected $i = 3$ indirect query results and took the average of the overlapping evaluations to compute the final score for each indirect query experiment. For direct query experiments, we sampled $j = 10$ judgments at temperature 1.0 and reported the fraction of *yes* responses as a final score.

Search engine labels The Bing search engine API[7] is used for searching for the candidate title string on the Web. Note that even with exact string match, some flexibility beyond capitalization and punctuation is allowed. A manual inspection of 120 random examples, given in Appendix C, finds the use of the search engine to be a reliable method for detecting hallucinations.

4.2 Quantitative metrics

First, Table 1 shows the rates of hallucination for the three models studied. As expected, references produced by the newer models (which achieve higher scores on other benchmarks [20]) also exhibit a higher grounding rate or, equivalently, a lower hallucination rate. While this is expected, it is a positive indicator of the validity of our approach of using search engine results to measure hallucination. (This is discussed further in Appendix C.)

Table 1: The hallucination rate (out of 1000 generated titles), as determined by ground-truth labels assigned using the Bing search API.

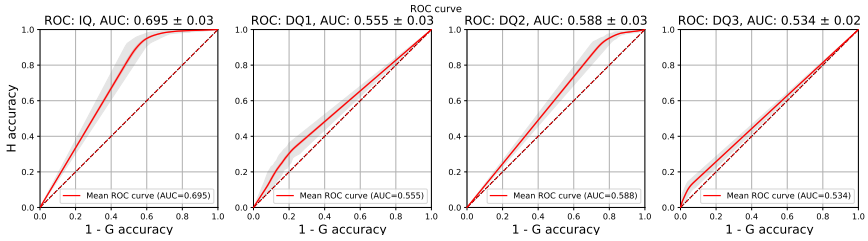
	H%
GPT-4	46.8%
ChatGPT	59.6%
GPT-3	73.6%

Since each of our querying strategies outputs a real-valued score, one can trade-off accuracy on G (i.e., how often truly grounded references are labeled G) and H (how often truly hallucinated references are labeled H) by thresholding the score to form a G or H classification. The standard receiver operating characteristic (ROC) curves based on these thresholded scores are shown for each approach and model in Figs. 5a, 5b, and 5c. These figures enable one to explore different points on this trade off for each classifier. For the GPT-3 and ChatGPT models, the IQ procedure performs best as quantified via the area under the ROC curve (AUC). For GPT-4 (Fig. 5c), both the IQ and DQ approaches work well for classifying hallucination and groundedness with the IQ (AUC: 0.878) and DQ1 (AUC: 0.887) performing the best. The performance of each procedure generally improves as the model size increases. For smaller models, where the procedures perform worst, others have found that users are less likely to believe the generated text due to its inaccuracy [16]. We additionally display 95% confidence bands for each ROC curve using 100 bootstrap replicates and a 95% confidence interval for the AUC using the DeLong et al. [5] estimate of AUC standard error.

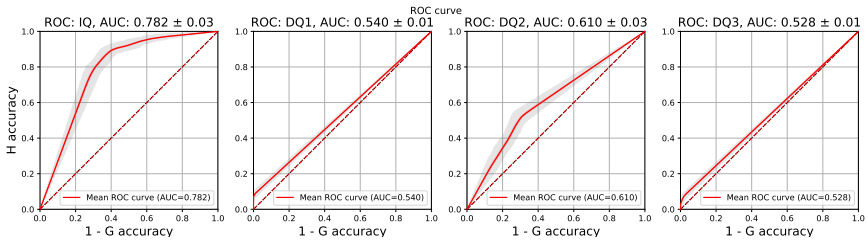
Each groundedness classifier can also be used as a filter to generate a list of likely grounded references for a literature review based on the raw generations of an LM. Aside from relevance, which we do not study in this work, two primary quantities of interest to a user of this filter would be the fraction of references preserved (more references provide a more comprehensive review) and the fraction of preserved references which are actually hallucinations. Fig. 7 shows how these two quantities can be traded off. As one varies the threshold of G/H classification and returns only those references classified as grounded, the false discovery rate (FDR) captures the fraction of references produced which are hallucinations. Users may have a certain rate of tolerance for hallucinations, and one would like to maximize the number of generated references subject to that constraint. For GPT-3 and ChatGPT, the IQ method achieves significantly lower FDR and a provides a substantially better FDR-preservation rate trade-off than the other approaches. For GPT-4, both IQ and DQ methods offer low FDR with comparable trade-offs. Fig. 7 also displays 95% FDR prediction intervals (lighter bands) computed from the quantiles of 100 bootstrap replicates and 95% confidence intervals (darker bands) for expected FDR computed from the bootstrap mean ± 1.96 times the bootstrap standard error.

Overall, our hypothesis that indirect queries would be more reliable than direct queries appears to hold for ChatGPT and GPT-3; for GPT-4 the direct queries were similarly effective. Finally, we now observe that one can improve accuracy for all models using a combination of direct and indirect queries.

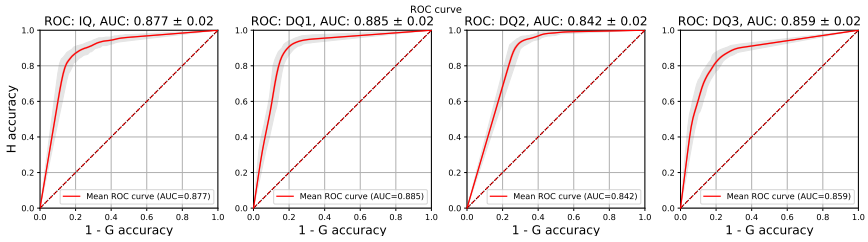
[7] https://www.microsoft.com/en-us/bing/apis/bing-web-search-api



(a) GPT-3



(b) ChatGPT



(c) GPT-4

Figure 5: ROC Curves for the four procedures, independent queries and direct queries 1-3, left to right. In (a) the procedures have little effect on mitigating hallucination in the GPT-3 model, where hallucination was rampant, though the IQ does help the most. In (b) again the IQ procedure does help while most of the DQ procedures are of little value. In (c), for GPT-4, all procedures are significantly effective, with large overlaps in the confidence intervals and AUCs.

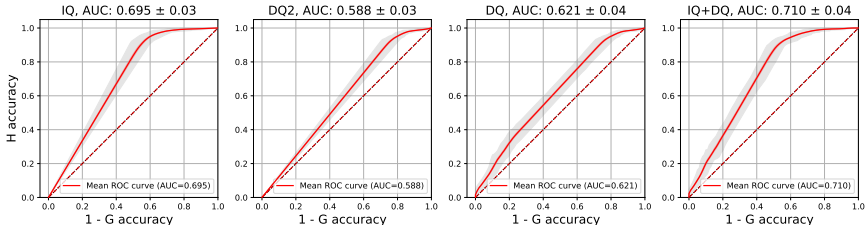
Ensemble of the approaches. We find that classification performance increases when we take ensemble of different approaches, as illustrated by ROC curves in Fig. 6. For creating the ensemble of the approaches, we simply compute the mean of the scores and use them as thresholds. The ensemble of three direct query approaches (computed using the equally-weighted mean of the DQ1, DQ2, and DQ3 scores), which we refer to as simply *DQ*, performs slightly better than the best performing direct query approach. The ensemble of IQ and DQ (computed using the 50-50 mean of IQ and the DQ mean), referred to as *IQ+DQ* performs the best for every model.

The compute costs, which involve ≈6.6 million tokens and $412, are discussed in Appendix B.

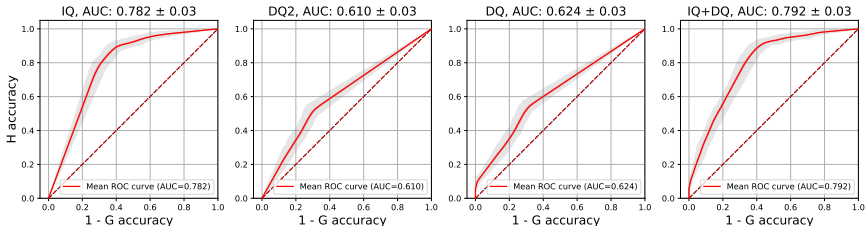
4.3 Qualitative findings

A qualitative examination of the titles generated by the LMs and their classifications according to the Bing search API revealed several interesting observations:

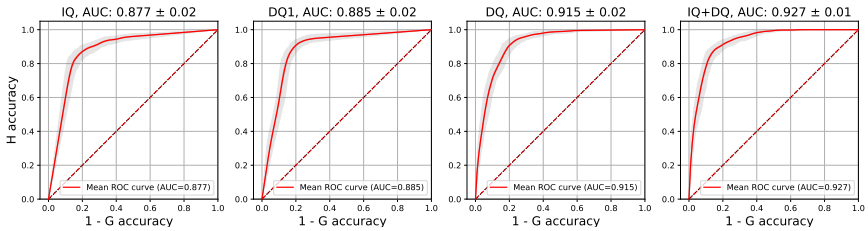
1. Many hallucinated titles were combinations of multiple existing titles.
2. The Bing quoted search heuristic is more lenient than exact match, ignoring more than just capitalization and punctuation. However, presumably since Bing quoted search is designed to facilitate title searches, it works well.
3. Some hallucinations were "plausible sounding" such as *A survey on X* for topic *X*, even when such a survey did not exist.
4. Direct methods may fail to identify hallucinations on "plausible sounding" titles such as surveys or book chapters. The indirect method also sometimes failed to identify a hallucination because



(a) GPT-3



(b) ChatGPT



(c) GPT-4

Figure 6: Ensembles combining approaches outperform the best single approach. Left to right: ROC curves for IQ, the best performing direct query approach, the ensemble DQ averaging the three direct query approaches, and the ensemble of IQ and DQ approaches. The ensemble of DQ approaches performs a bit better than the best performing DQ approach. The ensemble of IQ+DQ approaches performs the best for all models. For all three models, the biggest ensemble IQ+DQ performs best.

the LM would consistently produce a "likely author" based on the title, for a given non-existent paper. For example, GPT-4 hallucinated the title *Introduction to Operations Research and Decision Making*, but there is a real book called *Introduction to Operations Research*. In all three indirect queries, it hallucinated the authors of the existing book, *Hillier Frederick S., Lieberman Gerald J.*. Similarly, for the hallucinated title *Exploratory Data Analysis and the Role of Visualization*, 2 of 3 indirect queries produced *John W. Tukey*, the author of the classic, *Exploratory Data Analysis*.

5. The indirect method may sometimes fail to identify a grounded paper title which it can recognize/generate, as it may simply not be able to generate authors not encoded in its weights.

Since, in many applications, identifying potential hallucinations is more important than recognizing all grounded citations, errors due to falsely marking an H as a G are arguably more problematic than classifying a G as an H. A manual examination of 120 examples is given in Appendix C.

5 Conclusions, Limitations, and Future Work

This work investigates the hallucinated reference problem in LMs and provides a methodology by which LMs can be used for self-detection of hallucinations. Both direct and indirect queries were found to be effective for language models, and combining multiple methods led to further improvements in accuracy.

There are several limitations of this work. First, as discussed earlier, because we used LMs with inaccessible training data, we cannot conclude what is truly grounded versus hallucination. Second,